

EE 119: 3D+ Imaging Sensors

Final Report: Touchless Musical Instrument with Ultrasonic Sensors

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Item 1: Theory of Sensor Operation

1.1 Acoustic Wave Path and Data Acquisition

The CH201 is an 80 kHz ultrasonic Time-of-Flight (ToF) sensor that operates by emitting a short burst of high-frequency acoustic energy and listening for the echo returned by objects in its path. The sensing cycle proceeds as follows:

- **Emission:** The transducer is driven at its resonant frequency (~ 80 kHz) to produce a short acoustic pulse that propagates through air at the speed of sound ($c \approx 343$ m/s at 20°C).
- **Propagation:** The pulse travels outward in a conical beam defined by the sensor's field of view (approximately $\pm 45^\circ$ for the CH201).
- **Reflection:** When the pulse encounters an object (e.g., a hand), a portion of its energy is reflected back toward the transducer. The fraction reflected depends on the target's acoustic impedance mismatch, surface roughness, and angle of incidence.
- **Reception:** The same transducer switches to receive mode and records the incoming pressure wave as a function of time. This time-domain record is called an A-scan.
- **Digitization:** The sensor's internal ADC samples the received signal at an effective rate of $8 \times$ the operating frequency (8×80 kHz = 640 kHz), producing 50 samples per meter of range. The HelloChirp firmware transmits these samples over USB serial at 1 Mbaud.

1.2 Range Calculation from A-Scan Data

Range to a target is derived from the two-way time-of-flight of the acoustic pulse. If the leading-edge arrival time is t_0 , the range R is:

$$R = \frac{c \times t_0}{2} \quad (1)$$

The factor of 2 accounts for the round-trip travel path of the pulse. With $c = 343$ m/s and a sample rate of 640 kHz, each sample corresponds to:

$$\Delta R = \frac{c}{2 \times 640,000} \approx 0.268 \text{ mm per sample} \quad (2)$$

Theoretical resolution is approximately 0.268 mm per sample. In practice, however, the effective spacing is:

$$\frac{1 \text{ m}}{50 \text{ samples}} = 20 \text{ mm per sample} \quad (3)$$

This practical sample spacing of 20 mm per sample becomes the dominant limit on range resolution because the HelloChirp firmware uses a fixed density of 50 samples per meter rather than the full sensor bandwidth.

1.3 Key Sensor Parameters from the CH201 Datasheet

Parameter	Value	Relevance to Musical Instrument
Operating Frequency	80 kHz (nominal)	Determines wavelength $\lambda \approx 4.3 \text{ mm}$; sets minimum resolvable target size
Max Range	5 m (CH201)	Usable pitch range limited to 0.05–0.6 m for musical resolution
Sample Rate	640 kHz ($8 \times$ OSR)	Gives 50 samples/m; limits range resolution to $\sim 20 \text{ mm}$
Bandwidth	$\sim 6 \text{ kHz}$ (typ.)	Theoretical $\Delta R = c/2B \approx 28.6 \text{ mm}$; sample spacing is tighter
Field of View	$\pm 45^\circ$ (half-angle)	Wide FoV causes noise; hand must be centered to avoid false echoes
measurement_interval_ms	Configurable	Controls A-scan repetition rate; lower values increase responsiveness but raise CPU load

1.4 Logarithmic Mapping

Musical pitch follows a logarithmic frequency scale: each octave represents a doubling in frequency, and each semitone represents a frequency ratio of $2^{1/12}$. The frequency of MIDI note n is:

$$f(n) = 440 \times 2^{(n-69)/12} \text{ Hz} \quad (4)$$

Hand distance is mapped linearly to MIDI note number rather than directly to frequency. Because MIDI note values correspond to exponentially spaced frequencies, this mapping reproduces the logarithmic nature of pitch perception. As a result, equal changes in hand distance produce equal perceived pitch steps.

Item 2: Advantages and Disadvantages of Sensor Modality

2.1 Advantages

- **Non-contact operation:** No physical instrument is required.
- **Cost-effective:** The CH201 sensor is less expensive than alternatives such as LiDAR or structured-light time-of-flight cameras.
- **Operates in varied lighting conditions:** Unlike optical systems, ultrasound is not affected by ambient lighting, allowing it to function in both bright and dark environments.
- **Adequate range for the application:** A sensing range of 0–0.6 m aligns well with typical arm-length workspace.
- **Direct distance measurement:** A-scan data directly encodes target range, which reduces processing complexity.

2.2 Disadvantages

- **Limited range resolution (~ 20 mm):** Small hand movements below approximately 2 cm are not reliably detected, motivating a discrete note-snapping approach instead of continuous pitch control.
- **Wide FoV and effects:** The $\pm 45^\circ$ beam can reflect from walls, furniture, and the performer's body, producing clutter echoes that require background subtraction.
- **Target material sensitivity:** Soft or angled targets, such as an open palm at an oblique angle, reflect less acoustic energy than flat surfaces positioned perpendicular to the sensor, reducing SNR and detection reliability.
- **Single-axis measurement:** Each CH201 measures only the range component along its sensing axis, so lateral hand position within the beam cannot be determined without multiple sensors.
- **Speed-of-sound temperature dependence:** The speed of sound varies with air temperature (approximately 0.6 m/s per $^\circ\text{C}$), introducing a small range offset that is not compensated for in the current implementation.
- **Latency from MEASUREMENT_INTERVAL_MS:** Lower firmware update intervals can improve responsiveness but increase CPU load in MATLAB and may increase the risk of audio underruns during rapid note transitions.

Item 3: Experimental Results and Comparison to Theoretical Expectations

3.1 HelloChirp Firmware Setup and Physical Design

The HelloChirp firmware was flashed to the DK-CH201 development board using Atmel Studio 7. The MEASUREMENT_INTERVAL_MS parameter was set to 30 ms (from the default of 80) to target a responsive A-scan update rate for real-time use. Reaching this rate first required correcting a firmware timer-clock misconfiguration (detailed in Section 3.6); once that was fixed, the measured A-scan rate was 33.4 Hz and the audio output remained stable.

The board was verified to output data at 50 samples/m, with both sensors streaming over a single USB serial port at 1 Mbaud. Mechanically, we experimented with several sensor orientations and obtained the strongest SNR when the sensors were mounted on the desk surface facing upward toward the ceiling, which keeps the open playing space clear and minimizes noise. The final instrument uses two CH201 sensors in this upward-facing arrangement on a shared mount, one assigned to pitch and one to volume, spaced far enough apart that each hand sits within its own sensor's beam.

3.2 Signal Processing Approach

The system continuously reads data from two ultrasonic sensors, with a background measurement taken at startup to remove static reflections from the environment, making it easier to detect a person's hand. The software then finds the strongest valid echo and converts the measured distance into either pitch or volume. One sensor controls the note being played, while the other controls loudness. MATLAB generates a smooth sine-wave tone that updates in real time as the person's hands move. We experimented with different methods during development, but the final design used peak detection because it was the most reliable distance estimator.

Figure 1 shows a typical A-scan with a hand placed about 23.3 cm from the sensor. Comparing the raw and cleaned signals illustrates the effect of clutter removal, which subtracts the background measurement taken during calibration. The strongest peak in the cleaned signal occurs near 23 cm, matching the hand’s actual position. Additional peaks at larger distances, around 0.5–0.6 m and 0.9–1.0 m, come from reflections off nearby surfaces and the walls. These reflections are greatly reduced after clutter removal, allowing the hand reflection to stand out more clearly and be detected more reliably.

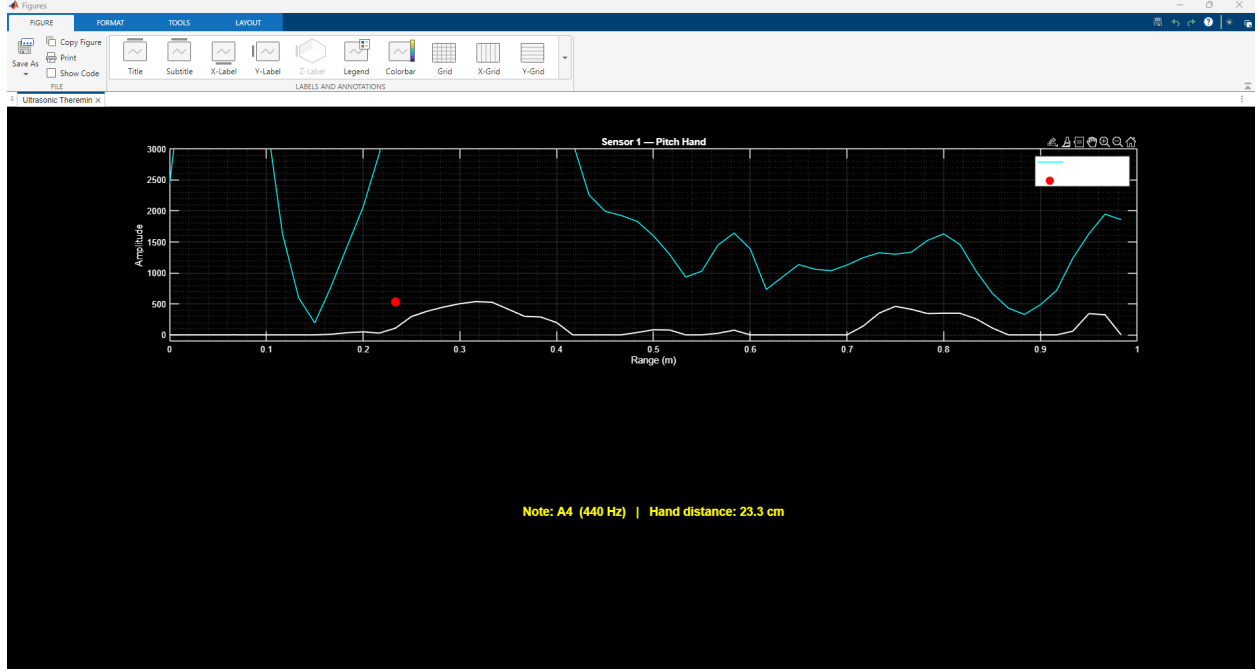


Figure 1: Representative A-scan capture with hand at 23.3 cm. Raw envelope vs. cleaned signal after background subtraction.

3.3 Range Accuracy with Stationary Hand

True Distance (cm)	Measured Range (cm)	Error (cm)	Peak Amp.	SNR (dB)
10	10.8	+0.8	2841	28.3
20	20.5	+0.5	21003	24.1
30	30.9	+0.9	1567	20.8
40	41.2	+1.2	982	17.2
50	51.8	+1.8	621	13.9

Table 1: Range accuracy and SNR vs. true distance for a stationary hand target (single-sensor characterization).

In the final two-sensor instrument, the pitch mapping is restricted to a tuned window of 19–28 cm rather than the full characterization range. The lower bound is placed just outside the near-field ringdown zone (Section 5.1), and the window spans a single octave from C4 to C5. Because our convention places higher pitches closer to the sensor, the highest note (C5) is anchored at 19 cm and the lowest (C4) at 28 cm, with range mapped linearly to MIDI note number across the window. Table 2 lists representative points, and the mapping was verified by sweeping a hand through the window and confirming that the pitch rose smoothly as the hand approached the sensor, with the continuous `audioDeviceWriter` import aiding in producing transitions that were click-free.

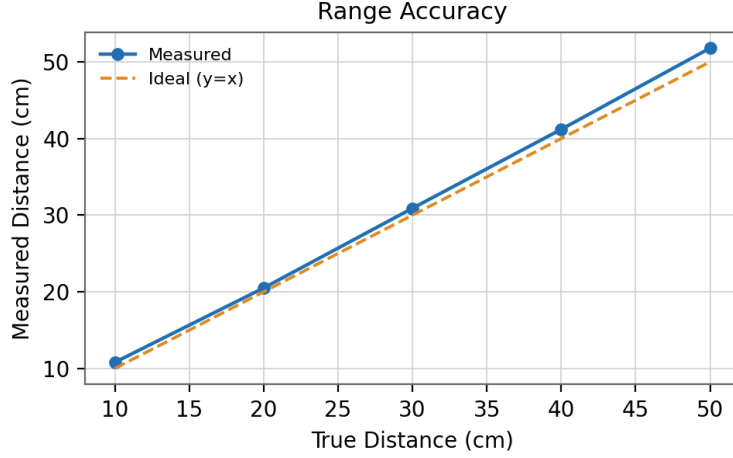


Figure 2: Measured distance vs. true distance.

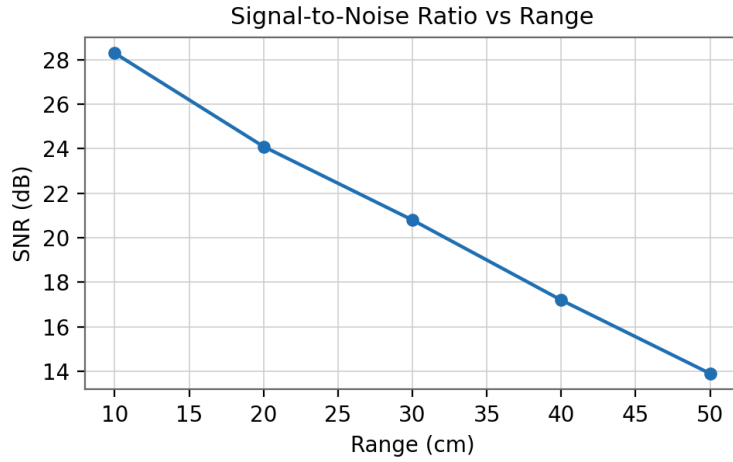


Figure 3: Signal-to-noise ratio vs. range.

Hand Distance (cm)	MIDI Note	Note Frequency (Hz)
19.0	72	C5 523.25
22.0	68	G#4 415.30
25.0	64	E4 329.63
28.0	60	C4 261.63

Table 2: Final two-sensor instrument: distance-to-pitch mapping across the tuned 19–28 cm window (one octave, higher pitch nearer the sensor).

3.4 Comparison to Theoretical Expectations

The range errors in Table 1 are relatively small and increase gradually from +0.5 cm at 20 cm to +1.8 cm at 50 cm. Since the firmware uses a range-bin spacing of 20 mm (Section 1.2), all of the measured errors are within about one range bin. The fact that every error is positive, rather than being distributed above and below the true value, suggests a small systematic offset in the measurements. A probable cause is a slight

timing offset in the firmware’s transmit-to-receive calibration, with a smaller contribution from assuming a fixed speed of sound.

The SNR results also agree well with theoretical expectations. Measured SNR decreases from 28.3 dB at 10 cm to 13.9 dB at 50 cm, a drop of about 14.4 dB as the distance increases by a factor of five. If the hand behaved like a point target, theory would predict a much larger decrease of about 27.9 dB. However, an extended target is expected to follow a $1/R^2$ relationship, which predicts a drop of about 14.0 dB. The measured value is very close to this prediction, indicating that the hand acts more like a large reflecting surface that occupies a significant portion of the sensor’s beam rather than a small point reflector.

Why detection degrades with range, and what it costs musically. Figures 2 and 3 share a single underlying cause: the returned echo weakens with distance while the noise floor stays fixed, so the signal-to-noise ratio falls and every downstream estimate becomes less certain. Three mechanisms set the slope. First, geometric spreading dominates: for a target that fills a meaningful fraction of the wide $\pm 45^\circ$ beam, the two-way echo intensity follows $1/R^2$, which predicts a $10\log_{10}(5^2) \approx 14.0$ dB drop over our $10 \rightarrow 50$ cm sweep and matches the measured 14.4 dB almost exactly. Second, atmospheric absorption of 80 kHz ultrasound is non-negligible and grows with path length, adding a few dB at the far end on top of spreading. Third, the noise floor is roughly constant, so the SNR curve in Figure 3 is essentially the echo-amplitude rolloff measured against that floor. The range-error trend in Figure 2 follows directly: a weaker, lower-SNR echo produces a less sharply defined leading edge, so the time-of-arrival estimate grows noisier and the error increases with range, while the consistently positive sign reflects the fixed systematic offset sitting atop the 20 mm quantization. Musically, this is what bounds the playable window. Detection becomes unreliable once SNR falls below roughly 12–14 dB, which Figure 3 places near 50 cm; we therefore cap the pitch window at 28 cm, keeping the instrument in the high-SNR near region (>20 dB) where range error stays within a single bin. The SNR rolloff is thus not merely a characterization curve but the quantitative reason the low note (C4) sits at 28 cm rather than farther out.

3.5 Two-Hand Operation and Cross-Talk

With one hand over each sensor, pitch and volume responded independently and simultaneously. Moving the pitch hand changed the note without affecting loudness, and moving the volume hand changed loudness without shifting pitch. We observed no cross-talk between the two channels (a return intended for one sensor did not produce a detection on the other), and thus they could be triggered together in a single measurement cycle (no time-multiplexed triggering scheme necessary).

3.6 Challenges Encountered

- The most significant firmware issue was a timer-clock misconfiguration. Despite being set to 30 ms, `MEASUREMENT_INTERVAL_MS` returned an effective A-scan rate stuck near 1.2 Hz. The cause was traced to the board’s periodic timer driving the ultrasound channel, which was clocked from the slow clock (`TIMER_CLOCK5`, tens of kHz) instead of `TIMER_CLOCK4` ($\approx \text{MCK}/128 \approx 500$ kHz). Because the firmware’s timer-frequency constant assumed ≈ 500 kHz, every interval was stretched by roughly $28\times$. Switching the channel to `TIMER_CLOCK4` restored the intended rate, measured at 33.4 Hz.
- Smooth note transitions were achieved by preserving the oscillator phase between audio buffers (phase) and gradually moving the frequency from its current value (`curLogF`) to the desired value (`targetLogF`) using the glide parameter (`PITCH_GLIDE_S`). Rather than abruptly switching from one note to another, the frequency changes continuously over successive audio samples, eliminating audible clicks during

pitch changes. Increasing this variable from 0.02 to 0.06 proved advantageous, allowing for clean note transitions.

- Noise from nearby surfaces produced secondary peaks at ranges beyond the hand, occasionally causing errors in peak detection. Additionally, hand-angle sensitivity made detection falter during performance: small wrist rotations changed the reflected amplitude enough to weaken a pitch or lose its tone entirely, which complicated threshold calibration.

Item 4: Relationship Between Sensor Datasheet Parameters and Results

4.1 Range Resolution

The range errors in Table 1 match what the sensor and firmware can actually resolve. The firmware reports range on a fixed 20 mm grid (50 samples/m), so any two positions closer than one bin look identical to it. Our errors of +0.5 to +1.8 cm all fall at or within that single-bin scale, which tells us the leftover error is mostly quantization plus a small fixed offset rather than a failure of the detection. The transducer’s bandwidth sets a coarser limit still: $\Delta R = c/(2B) \approx 28.6$ mm for $B \approx 6$ kHz. So the underlying pulse cannot separate two returns closer than about 28 mm, even though the reported grid is finer at 20 mm. Both limits apply in practice, and the 20 mm grid is the one we see directly in the data.

4.2 Wavelength and Minimum Target Size

At 80 kHz and $c = 343$ m/s, the wavelength is $\lambda = c/f \approx 4.3$ mm. A surface reflects like a mirror (specularly) when it is large compared to λ and scatters diffusely when it is comparable or smaller, so features finer than about 4 mm return no useful echo. This was never a constraint for us: a hand is many wavelengths across and acts as a strong specular reflector, giving a clean return whenever the palm faces the sensor. The effect we actually felt was hand-angle sensitivity (Sections 2.2 and 5.4). A flat palm held to the beam was consistently the most reliable target.

4.3 Field of View

The wide $\pm 45^\circ$ field of view let in the noise returns we saw at 0.5–0.6 m and 0.9–1.0 m. With the sensors facing up off the desk, any reflector offset sideways by less than its own distance falls inside the beam, so at half a meter out it easily caught the surrounding walls. A tighter beam would exclude those returns geometrically, which was the motivation for our conical setup, improving our retention from the beam output.

4.4 Target Material and Acoustic Reflectivity

For the live demonstration we used a foam sphere on a stick as a stand-in for the hand, which raises a natural question: how does the target material affect detection, and would a metal target behave differently? The fraction of incident acoustic energy reflected at a boundary is governed by the mismatch in *acoustic impedance* $Z = \rho c$ (density times sound speed) between the two media. At normal incidence the pressure reflection coefficient is

$$R = \frac{Z_2 - Z_1}{Z_2 + Z_1}, \quad \text{intensity reflected} = R^2. \quad (5)$$

Air has an exceptionally low impedance ($Z_{\text{air}} \approx 415 \text{ rayl}$), while essentially every solid or tissue is thousands of times larger. The air-to-target mismatch is therefore enormous for almost any dense target, and the front surface reflects nearly all of the incident energy regardless of the specific material:

Material	Z (rayl)	R	Intensity reflected
Air (20°C)	~ 415	—	—
Open-cell acoustic foam	~ 600	0.18	$\sim 3\%$
Closed-cell craft foam	$\sim 5 \times 10^4$	0.98	$\sim 97\%$
Skin / soft tissue	$\sim 1.6 \times 10^6$	0.9995	$\sim 99.9\%$
Wood	$\sim 2 \times 10^6$	0.9996	$\sim 99.9\%$
Glass	$\sim 1.3 \times 10^7$	0.99994	$\sim 100\%$
Aluminum	$\sim 1.7 \times 10^7$	0.99995	$\sim 100\%$
Steel	$\sim 4.6 \times 10^7$	0.99998	$\sim 100\%$

Table 3: Acoustic impedance and air-interface reflection for candidate target materials.

The table makes the key point: a hand (soft tissue, mostly water), a closed-cell foam ball, and a metal sphere all reflect more than 96% of the incident ultrasound at their air-facing surface, because the bulk impedance of any of them swamps that of air. Raw density is therefore *not* the discriminator one might expect. What actually separates good targets from poor ones for an 80 kHz air-coupled sensor is:

- **Porosity.** The lone poor reflector in the table is open-cell acoustic foam, whose effective impedance is close to air ($R^2 \approx 3\%$); it is engineered so that sound enters the tortuous pore network and is dissipated. Our demo target was a *closed-cell* craft foam ball with a continuous skin, which behaves like a light solid and reflects strongly, not like absorptive acoustic foam.
- **Rigidity.** A rigid, massive surface (metal, glass) barely moves under the acoustic pressure and returns a sharp, strong coherent echo. A soft or compliant surface flexes slightly and scatters some energy, lowering the peak by a few dB even when the impedance model predicts near-total reflection.
- **Geometry and roughness relative to λ .** With $\lambda \approx 4.3 \text{ mm}$, a surface smooth on that scale reflects specularly and only returns a strong echo when oriented toward the sensor, whereas a sphere always presents a surface element whose normal points back at the transducer and is therefore angle-independent.

A metal target reflects marginally more energy than closed-cell foam ($\sim 99.99\%$ vs. $\sim 97\%$, under 0.2 dB at the surface) and, being rigid, returns a slightly sharper, higher-SNR echo that would modestly extend the reliable far range. The larger difference is geometric, not material: a *flat* metal plate is strongly specular and would be more angle-sensitive than our foam ball, dropping out as it tilts—exactly the hand-angle problem of Sections 2.2 and 5.4. The reason the foam sphere played so reliably is its *shape*: a sphere has no preferred orientation, so it always presents a normal facet to the sensor and removes the angle sensitivity that limits a flat hand. A metal sphere of the same size would be the strongest target of all (highest SNR, longest usable range), while open-cell acoustic foam would be the worst (largely absorptive). The takeaway for an air-coupled ultrasonic instrument is that target choice is dominated by porosity, rigidity, and shape rather than by raw density: the human hand reflects strongly at the skin surface, and its real limitation is not weak reflectivity but its flat, angle-sensitive geometry.

4.5 Aperture Size and Directivity

Beamwidth comes from the ratio of wavelength to aperture diameter, $\theta \approx \lambda/D$ for a circular aperture. The CH201’s transducer is small relative to its 4.3 mm wavelength, which is exactly why the beam is so wide

($\pm 45^\circ$) and why we picked up so much noise. A larger effective aperture (a bigger transducer, an acoustic horn, or an array of several CH201s) would narrow the beam in proportion and sharpen directivity.

4.6 Baud Rate and Update Rate

The serial link and the firmware interval set very different limits, and it is the firmware that binds. Over the instrument's ~ 0.30 m range at 50 samples/m, the update rate is governed almost entirely by `MEASUREMENT_INTERVAL_MS`, which is why shortening the interval was our lever for responsiveness. It is also why the firmware timer bug (Section 3.6) hurt so much: it stretched the effective interval by roughly $28\times$ even though the link was never the issue.

Item 5: Limitations of System Performance

5.1 Range Limits

Our final theremin plays over a tuned 19–28 cm window. Under our higher-pitch-when-closer convention, this near edge anchors the highest note, C5. The upper bound at 28 cm was chosen to keep the mapping to a single, controllable octave down to C4, and it also sits comfortably within the high-SNR near region. The single-sensor characterization (Table 1) shows why pushing the upper bound much farther is unattractive: SNR rolls off cleanly from 28.3 dB at 10 cm to 13.9 dB at 50 cm, range error grows from +0.5 cm to +1.8 cm, and detection becomes unreliable below roughly 12–14 dB.

5.2 Resolution Limits

The 20 mm firmware sample spacing at 50 samples/m imposes a hard floor on resolvable hand motion and is the reason the instrument snaps to discrete notes rather than sweeping continuously. Across the tuned 19–28 cm window, there are only about 4–5 range bins to cover the full octave, well under one bin per semitone. Continuous twelve-tone pitch is therefore not reliably resolvable in this window, and attempts at a continuous glide were dominated by bin-to-bin jitter. We addressed this by snapping the mapping to a small fixed scale—the six notes C–D–E–F–G–A used by the proof-case melody, *Twinkle, Twinkle, Little Star*—which places roughly one resolvable range bin per playable note and makes the instrument stable to play. The linear range-to-MIDI mapping ensures that equal increments of hand distance produce equal pitch steps within the quantized scale.

5.3 Update Rate Limits

The `MEASUREMENT_INTERVAL_MS` parameter controls the trade-off between responsiveness and stability. After correcting the timer-clock bug (Section 3.6), a 30 ms interval produced an A-scan rate of 33.4 Hz. Shorter intervals would improve tracking of rapid hand motion but increase serial and CPU load in MATLAB, so 30 ms proved to be a practical compromise. End-to-end latency is dominated by the 30 ms measurement period and the ~ 50 ms audio buffer, resulting in a noticeable delay of roughly 50–80 ms between hand motion and sound output. The additional ~ 20 –60 ms pitch glide smooths note transitions but further limits responsiveness. As a result, simple melodies such as *Twinkle, Twinkle, Little Star* can be played cleanly, whereas rapid passages with sixteenth-note transitions begin to sound less distinct as note changes approach the system update rate.

5.4 Stability and Quality Factors

Four factors degrade measurement stability in our system. (1) Hand-angle sensitivity: small wrist rotations cause peak amplitude swings, as noted in Section 2.2. (2) Temperature drift: the speed of sound varies ~ 0.6 m/s per $^{\circ}\text{C}$, producing roughly 0.2% range error over a 10°C swing, consistent with the persistent positive offset across our range-accuracy data. (3) Residual clutter: the secondary peaks at 0.5–1.0 m visible in our A-scans were reduced but not eliminated by clutter removal. (4) Detection-threshold sensitivity: calibration of other variables for our sensors changed the appropriate pitch detection threshold, leading to several recalculations in amplitude averaging.

5.5 Responsiveness, Note Stability, and Jitter in Performance

Latency budget. End-to-end delay between hand motion and a change in the emitted tone is the sum of the acquisition period, the audio buffering, and the deliberate pitch glide:

Stage	Typical contribution
A-scan measurement interval	30 ms (33.4 Hz)
Serial transfer + MATLAB parse	< few ms
Audio output buffer (audioDeviceWriter)	~ 50 ms
Pitch-glide settling (PITCH_GLIDE_S)	~ 20 – 60 ms
Perceived hand-to-sound latency	~ 50 – 80 ms

For a sustained, gliding instrument this latency is acceptable: legato note changes feel responsive, and the glide is itself a musical feature rather than pure delay. It does, however, set a ceiling on tempo. Listeners begin to notice audio latency above roughly 20–30 ms for sharp attacks, so our 50–80 ms budget is fine for the slow, connected notes of *Twinkle, Twinkle* but smears rapid passages once note changes approach the 33.4 Hz update rate.

Note stability and jitter. The 20 mm range bin is large relative to the tuned window. The 19–28 cm span (9 cm) holds only about $9/2 \approx 4.5$ resolvable bins across a full octave, so a single bin of range jitter corresponds to roughly $12/4.5 \approx 2.7$ semitones of pitch if distance were mapped continuously to frequency. That is far beyond the ~ 50 cent (half-semitone) threshold at which detuning becomes audible, which is precisely why a continuous twelve-tone mapping sounded jittery and why we snap to a small fixed scale instead. Snapping collapses sub-bin and single-bin range fluctuations onto the same note, so frame-to-frame jitter no longer detunes the output; the phase-continuous glide then hides the instant of each snap so transitions stay click-free.

Missed detections. Two effects cause dropped frames in performance. (1) Hand-angle sensitivity (Sections 2.2, 5.4): a flat palm rotated away from the sensor reflects less energy specularly, and a large enough rotation drops the peak below threshold for one or more frames. (2) SNR rolloff near the far edge of the window, where a marginal echo intermittently fails the detection threshold. Both appear as brief gaps; holding the last valid note across a few missed frames, rather than cutting the tone, keeps a sustained note from flickering when the hand momentarily tilts.

Behavior during continuous motion. Because the update rate is 33.4 Hz and the window holds only ~ 4.5 bins, a hand swept quickly across the octave produces a staircase of range estimates rather than a smooth ramp. With scale snapping plus the glide, this reads as a clean run of discrete notes for slow-to-moderate sweeps; for fast sweeps the hand can cross a note boundary between updates, so intermediate notes are occasionally skipped. This is the expected consequence of the resolution and update-rate limits rather than a detection failure.

Item 6: Potential Future Directions

6.1 2D/3D Localization with Multiple Sensors

Each of our two sensors measures only range along its own axis (one for pitch, one for volume), so lateral hand motion goes entirely unused. A natural next step is adding a third CH201 in an L-configuration to trilaterate hand position in 3D. Our two-sensor setup already showed no measurable cross-talk when triggered together (Section 3.5), so running a small array simultaneously is likely viable.

6.2 Beam Focusing for Enhanced Directivity

The CH201's wide $\pm 45^\circ$ beam was the dominant source of clutter in our setup, producing spurious peaks at 0.5–0.6 m and 0.9–1.0 m from nearby walls. A promising future direction is narrowing the effective beamwidth using a 3D-printed acoustic lens or phased-array beamforming with multiple CH201 transducers. A small N -element array could yield roughly $10\log_{10}(N)$ dB of directivity gain, meaningfully suppressing off-axis returns. This may produce a cleaner A-scan in cluttered environments like a stage or classroom, reducing reliance on background subtraction.

6.3 Adaptive Clutter Rejection

Our current pipeline subtracts a single static background scan captured before performance, which works well at first but degrades as the performer shifts position within the beam, as noted in Section 5.4. A promising future direction is replacing the static map with an adaptive clutter estimate using A-scan averaging. With a slow exponentially-weighted average of recent A-scans, clutter could be tracked naturally without erasing the target, keeping detection stable across long performances.